

# INTRODUCTION TO APPLIED NUMERICAL METHODS

## Numerical Integration: Simpson's 1/3 Rule

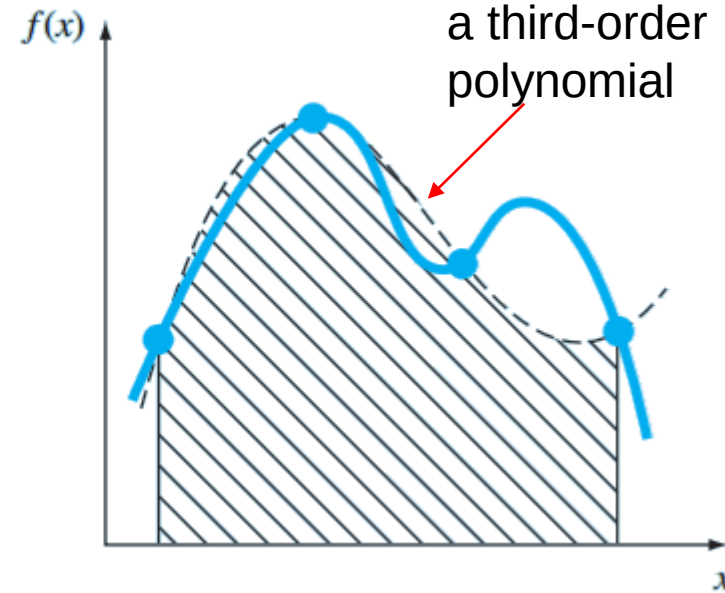
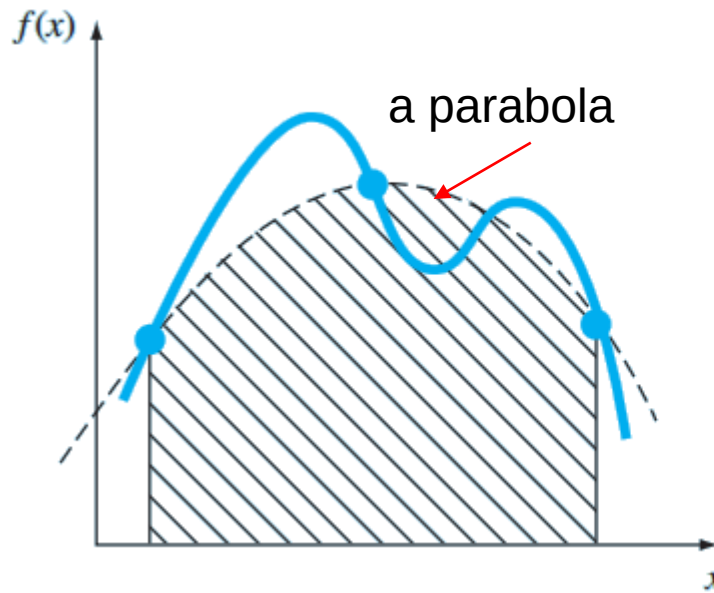
Lang Yuan, Associate Professor



# SIMPSON'S RULE

## Simpson's Rule

- Using high-order polynomial to connect points for improved accuracy
- A “parabola” can connect three data points
- A “third-order polynomial” can connect four data points
- Taking the integrals under these polynomials

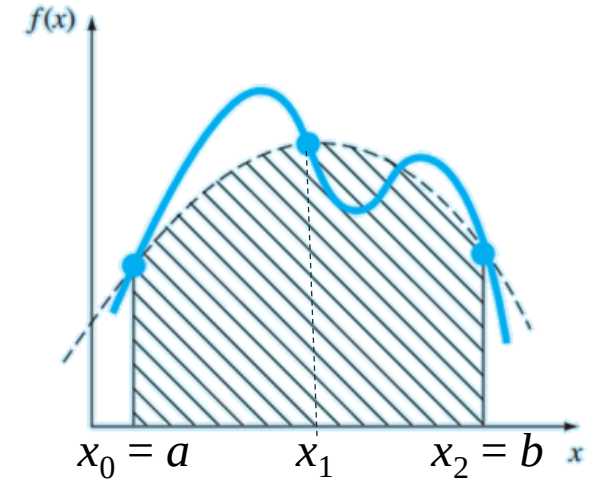


# SIMPSON'S 1/3 RULE

## The Simpson's 1/3 Rule:

- Using a “second-order polynomial” as the approximating function

$$I = \int_a^b f(x) dx \cong \int_a^b f_2(x) dx$$



- A 2<sup>nd</sup>-order **Lagrange** polynomial for  $f_2(x)$  within  $x = [a, b]$

$$f_2(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}f(x_0) + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}f(x_1) + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}f(x_2)$$

$$I = \int_{x_0}^{x_2} \left[ \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}f(x_0) + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}f(x_1) + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}f(x_2) \right] dx$$

$$I \cong \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)]$$

“1/3” rule

$$h = (b - a)/2$$



$$I \cong \underbrace{(b - a)}_{\text{Width}} \underbrace{\frac{f(x_0) + 4f(x_1) + f(x_2)}{6}}_{\text{Average height}}$$

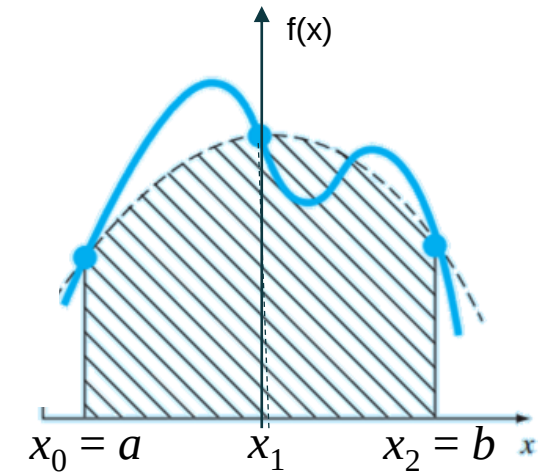
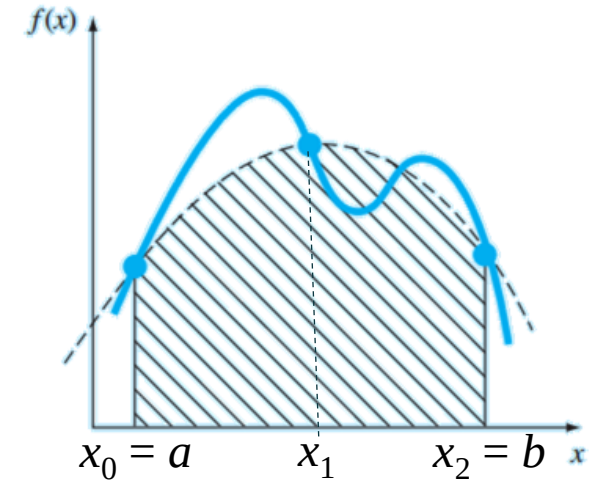
# SIMPSON'S 1/3 RULE: HOW TO PROVE IT?

$$I = \int_a^b f(x) dx \cong \int_a^b f_2(x) dx$$

$$f_2(x) = a_0 + a_1x + a_2x^2$$

Shifting the y axis to the middle point:

$$\begin{aligned} \int_a^b f_2(x) dx &= \int_{-h}^h f_2(x) dx = \left( a_0x + \frac{a_1x^2}{2} + \frac{a_2x^3}{3} \right) \Big|_{-h}^h \\ &= 2a_0h + \frac{a_1(h^2 - (-h)^2)}{2} + \frac{a_2(h^3 - (-h)^3)}{3} \\ &= 2a_0h + \frac{2a_2h^3}{3} \end{aligned}$$



# SIMPSON'S 1/3 RULE: HOW TO PROVE IT? – METHOD 2

$$\int_{-h}^h f_2(x)dx = 2a_0h + \frac{2a_2h^3}{3} = \frac{h}{3}(6a_0 + 2a_2h^2)$$

$$f_2(h) = a_0 + a_1h + a_2h^2 = f(x_2) \quad (1)$$

$$f_2(-h) = a_0 - a_1h + a_2h^2 = f(x_0) \quad (2)$$

$$f_2(0) = a_0 = f(x_1) \quad (3)$$

Equation 1 +2:

$$f(h) + f(-h) = 2a_0 + 2a_2h^2$$

$$2a_2h^2 = f(h) + f(-h) - 2a_0$$

Therefore:

$$\int_{-h}^h f_2(x)dx = \frac{h}{3}(6a_0 + 2a_2h^2) = \frac{h}{3}(4a_0 + f(h) + f(-h)) = \frac{h}{3}(f(x_0) + 4(f(x_1) + f(x_2)))$$

# AN EXAMPLE

Using the Simpson's 1/3 rule to numerically integrate

$$f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$$

From  $a = 0$  and  $b = 0.8$  (the exact value of integral is 1.640533).

**Solution:**

$$I \cong \underbrace{(b - a)}_{\text{Width}} \underbrace{\frac{f(x_0) + 4f(x_1) + f(x_2)}{6}}_{\text{Average height}}$$

$n = 2$  (3 data points),  $h = 0.4$

$$x_0 = a = 0 \quad f(0) = 0.2$$

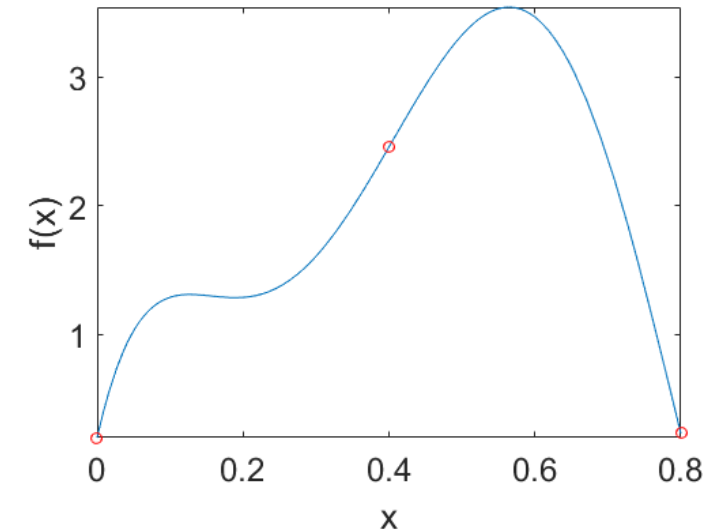
$$x_2 = b = 0.8 \quad f(0.8) = 0.232$$

$$x_1 = 0.4 \quad f(0.4) = 2.456$$

$$I \cong 0.8 \frac{0.2 + 4(2.456) + 0.232}{6} = 1.367467$$

$$E_t = 1.640533 - 1.367467 = 0.2730667$$

$$\varepsilon_t = 16.6\%$$



# MULTIPLE-APPLICATION SIMPSON'S 1/3 RULE

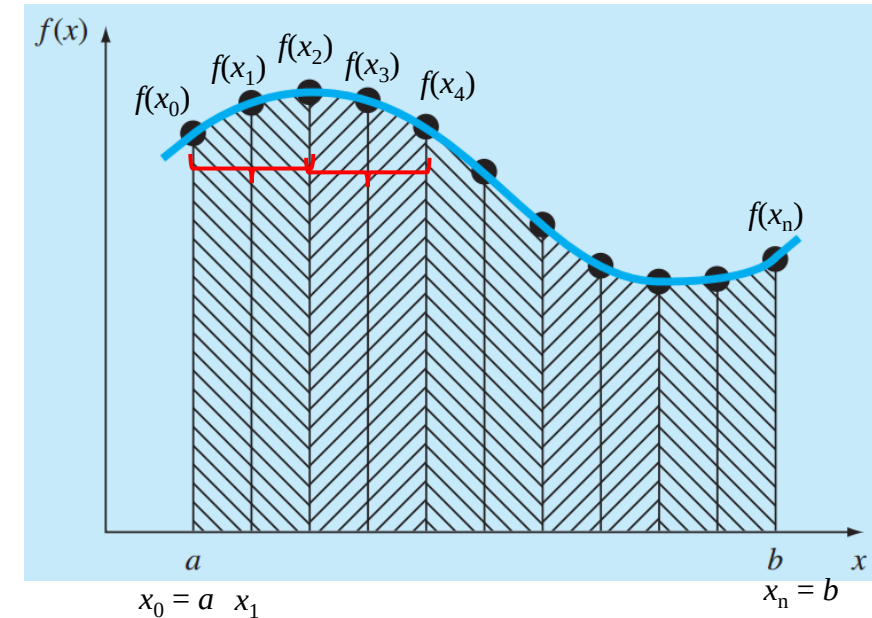
## Improving the Single Application of the Simpson's Rule:

- Divide the integration interval into a number of segments/sub-intervals
- Apply the Simpson's 1/3 rule to every 2 segments

$$h = \frac{b - a}{n}$$

$$I = \int_{x_0}^{x_2} f(x) dx + \int_{x_2}^{x_4} f(x) dx + \cdots + \int_{x_{n-2}}^{x_n} f(x) dx$$

$$I \cong 2h \frac{f(x_0) + 4f(x_1) + f(x_2)}{6} + 2h \frac{f(x_2) + 4f(x_3) + f(x_4)}{6} \\ + \cdots + 2h \frac{f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)}{6}$$



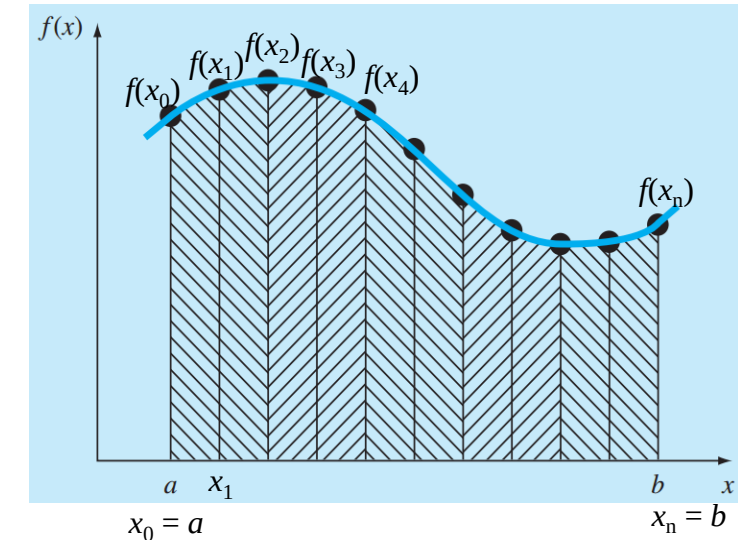
# MULTIPLE-APPLICATION SIMPSON'S 1/3 RULE

## Improving the Single Application of the Simpson's Rule:

- Divide the integration interval into a number of segments/sub-intervals
- Apply the Simpson's 1/3 rule to every 2 segments

$$I \cong 2h \frac{f(x_0) + 4f(x_1) + f(x_2)}{6} + 2h \frac{f(x_2) + 4f(x_3) + f(x_4)}{6} + \dots + 2h \frac{f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)}{6}$$

$$I \cong \underbrace{(b - a)}_{\text{Width}} \underbrace{\frac{f(x_0) + 4 \sum_{i=1,3,5}^{n-1} f(x_i) + 2 \sum_{j=2,4,6}^{n-2} f(x_j) + f(x_n)}{3n}}_{\text{Average height}}$$



**The segment number is “even” number**  
**Data point number: odd number**

**Coefficient “4”:** The odd-numbered points represent the middle term in each application

**Coefficient “2”:** The even-numbered points are shared by adjacent application (counted twice)

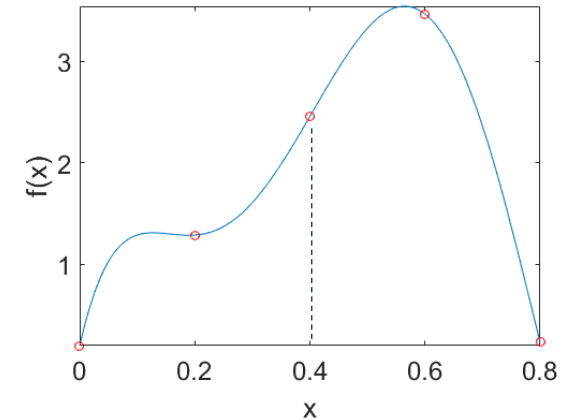


# AN EXAMPLE

Using four segments ( $n = 4$ ) and multiple applications of Simpson's rule to numerically integrate

$$f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$$

From  $a = 0$  and  $b = 0.8$  (the exact value of integral is 1.640533).



## Solution:

$n = 4$  (5 data points),  $h = 0.2$

$$f(0) = 0.2 \quad f(0.2) = 1.288$$

$$f(0.4) = 2.456 \quad f(0.6) = 3.464$$

$$f(0.8) = 0.232$$

$$I \cong \underbrace{(b - a)}_{\text{Width}} \underbrace{\frac{f(x_0) + 4 \sum_{i=1,3,5}^{n-1} f(x_i) + 2 \sum_{j=2,4,6}^{n-2} f(x_j) + f(x_n)}{3n}}_{\text{Average height}}$$

$$I = 0.8 \frac{0.2 + 4(1.288 + 3.464) + 2(2.456) + 0.232}{12} = 1.623467$$

$$E_t = 1.640533 - 1.623467 = 0.017067 \quad \varepsilon_t = 1.04\%$$

Using multi-segment Simpson's 1/3 rule significantly improves the integration accuracy